Small primitive attachments through a decorated tail

Human Compression project: bounded L044 proof component

This note proves the attachment exclusion in Proposition 5.11 of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, physical page 26. Its sixteen small markings are supplied by the separately accepted exact Table 2 classification. The source's alternating-attachment idea and its attribution to Fernandes–Leemans–Mixer are retained. The kernel-orbit argument below treats the extra central decorations explicitly; it does not assert IP for a projected tail. No whole-original or new-question claim is made.

Lemma 1 (Decorated alternating attachment)**.** *Let W, Y, R be disjoint finite sets, $Y = \{u, v, w, z\}$, and $c \in W$. Let T preserve all three sets and be transitive on W. Suppose $K = \ker(T \to S_Y)$ has a W-orbit O containing c with $|O| \geq 2$. Let a, b be involutions preserving $W \cup Y$ and R, with*

$$a|_{W\cup Y} = (c\ u)(v\ w), \qquad b|_{W\cup Y} = (u\ v)(w\ z).$$

If the action of K on R centralizes $a|_R$, then $\langle T, a, b\rangle$ contains the full alternating group on $W \cup Y$, acting trivially on R.

Proof. For every $j \in O$, conjugation by an element of K supplies

$$a_j = (j\ u)(v\ w)\, a|_R.$$

Products $a_j a_l$ for distinct j, l cancel the decoration and give pure three-cycles through u. They generate $A_{O\cup\{u\}}$, acting trivially elsewhere. Its conjugate by b is $A_{O\cup\{v\}}$. Alternating support gluing on their intersection O, of size at least two, gives $A_{O\cup\{u,v\}}$. Conjugating this by a and gluing gives $A_{O\cup\{u,v,w\}}$; conjugating that by b and gluing gives $A_{O\cup Y}$. All these are pure supported alternating groups: conjugating a permutation supported in $W \cup Y$ cannot introduce support in the invariant set R.

The normal subgroup K has its W-orbits permuted transitively by T. Conjugates of $A_{O\cup Y}$ therefore give $A_{O'\cup Y}$ for every such orbit O'. Their common set Y has four points, so repeated alternating support gluing gives $A_{W\cup Y}$ with trivial action on R. The gluing identity is $(p\ q\ x)^{-1}(p\ q\ y) = (p\ x\ y)$ in right-action notation, for distinct p, q in the common support. Thus no classification of T or IP assumption on it is needed. □

Lemma 2 (The twelve literal attachment templates)**.** *Take a marking $H = A_X$ from rows 4–16 of source Table 2, in either its printed order or its full reverse. Write its rank as d, $a = g_{d-1}$, $b = g_{d-2}$ and $H_- = \langle g_0, \ldots, g_{d-2}\rangle$. Either H_- fixes no point of X, or it fixes exactly one point c and the following properties hold. There are four distinct points u, v, w, z, all different from c, such that on $\{c, u, v, w, z\}$*

$$a = (c\ u)(v\ w), \qquad b = (u\ v)(w\ z).$$

Put $Y = \{u, v, w, z\}$, $R = X \setminus (\{c\} \cup Y)$, and $C = C_{S_{X\setminus\{c\}}}(H_-)$. Both a, b preserve R; C preserves Y, R and has order at most two. The group induced on R by a, b, C has order at most two and is even. The action of C on Y is trivial when $|X| = 11$. Finally $|\langle a, b\rangle|$ is ten or twenty, hence below sixty.

Proof. This is a finite check of the literal drawings, independent of the marking census. The mandatory `source_table2.json` is the exact previously matched edge transcription. `templates.g` checks each of the 26 orientations. It computes the entire H_- fixed set and the entire centralizer C, sets

$$u = c^a, \quad v = u^b, \quad w = v^a, \quad z = w^b,$$

and verifies the displayed restrictions, all invariant sets, the auxiliary action, its parity and the last-pair order. There are fourteen orientations without an endpoint and twelve with the stated template. Their exact vertex lists and complete centralizer elements are in the fresh receipt.

The positive orientations are rows 4 and its reverse; rows 5, 6,7,10; row 11 and its reverse; row 12; row 14; and row 16 and its reverse. Only rows 5,6,7 have a possibly nontrivial auxiliary group on R. In rows 6 and 7 its generator is the even involution (4 8)(5 9) in the literal upper-then-lower vertex numbering. This decoration is retained, not discarded by an unproved projection argument. The whole template program and data are required finite premises; the source is not used as an assertion that these tests pass. □

Theorem 3 (Primitive prefix exclusion)**.** *Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$ be transitive, with a connected string C-marking and all maximal omissions intransitive, where $n \ge 12$. Suppose $H = \langle \rho_0, \dots, \rho_{d-1} \rangle$ has nonempty support X and acts primitively and transitively there, with $m = |X| < 12$ and $d \ge (m-1)/2$. Then its coloured graph is row 1,2 or 3 of source Table 2, or the appropriate full reverse. The dual suffix assertion follows by reversing the whole marking.*

Proof. The prefix inherits IP, evenness and connectedness. An omission of one of its generators cannot remain transitive on X: otherwise the corresponding global omission connects every pair in X, so adding the omitted permutation, supported on X, cannot change its orbits. This contradicts global transitivity. The accepted Table 2 theorem therefore applies.

Suppose its row is among 4–16, so $H = A_X$ and $m = 9, 10$ or 11. Since X is proper and G is transitive, $d < r$. Any permutation commuting with H fixes X pointwise. Indeed it preserves the nontrivial H-orbit X, and the centralizer of the natural alternating action is trivial: the point stabilizer fixes only that point. Thus every ρ_j with $j \ge d+1$ fixes X pointwise. The only possible exit from X is $x = \rho_d$, which centralizes H_-. If $t \in X$ is sent outside, then for every $h \in H_-$, $(t^h)^x = (t^x)^h = t^x$, so $t^h = t$. The no-endpoint templates are impossible. In every remaining case the only exit point is the unique c, and x interchanges it with an outside point.

Let Y, R be its template and put $W = (\Omega \setminus X) \cup \{c\}$, $T = \langle \rho_d, \dots, \rho_{r-1} \rangle$. The nonfixed set of H_- is $X \setminus \{c\}$, so x preserves it and restricts there to an element of C. Consequently T preserves W, Y, R, has image of order at most two on Y, and its action on R belongs to the same even group of order at most two as $a|_R, b|_R$. In particular it centralizes $a|_R$.

The tail T is transitive on W. To see this, use connectivity of the full coloured permutation graph. All old prefix edges are inside X, and a tail edge can leave X only at c. Every excursion through X of a walk between points of W can therefore be deleted at c; the remaining walk uses only tail colours in W.

Let $K = \ker(T \to S_Y)$. Its normal orbits on W have equal size and there are at most two. For $m = 9$, $|W| \ge 4$; for $m = 10$, $|W| \ge 3$, and a partition into two equal orbits cannot have singleton parts. For $m = 11$, the template action on Y is trivial, so $K = T$ and $|W| \ge 2$. Thus in every case a K-orbit containing c has at least two points. Lemma 1 yields

$$A_{W \cup Y} \;\le\; J := \langle \rho_{d-2}, \dots, \rho_{r-1} \rangle,$$

with trivial action on R. Intersecting with $H = A_X$ gives $A_{\{c\} \cup Y} \cong A_5 \le H \cap J$. But full marked IP requires $H \cap J = \langle a, b \rangle$, of order at most twenty. This contradiction excludes every row 4–16. □

The source's intermediate groups. The argument also justifies the group shapes used in its case summary. All actions on R in the templates are even, so, since $G \leq A_n$, every element of J induces an even permutation on $W \cup Y$. We proved that the pure alternating factor is contained in J. Hence $J = A_{W \cup Y}$ if the auxiliary action is trivial, and otherwise $J = A_{W \cup Y} \times C_2$. This is an actual direct product on disjoint supports. The latter possibility occurs only for rows 5,6,7. The support size is $n - m + 5$, giving the source's $A_{n-4}, A_{n-5}, A_{n-6}$ factors respectively. This argument retains every decoration and does not infer a marked isomorphism from an abstract group order.

Scope and dependencies. The complete Table 2 packet is a paid premise. The whole new finite dependency is `templates.g`, `check.py` and the exact source graph data; GAP permutation arithmetic is retained. No subgroup census, source extraction or Lean rebuild is repeated. The source's subsequent supported-interval classification and whole L044 remain separate obligations.